

PC Network Summary

My network plan is built using the same principles I use to VR fashion systems: structure, clarity, purposeful layering, and long-term scalability. Techopoly Inc. is growing into a new 50,000-square-foot headquarters. The objective is to offer a high-performance, well-organized, and secure network that feels as smooth and dependable as a well-designed interface that is straightforward on the outside but built on a solid, fundamental architecture.

Three corporate servers, SRV-APP01, SRV-DB01, and SRV-DC01, each having fiber NICs, are supported by a 4-port fiber switch that is uplinked from the core switch. For critical workloads, including database operations, Microsoft Active Directory/DNS services, and application hosting, using multimode fiber ensures high-bandwidth, low-latency performance. To preserve uptime and avoid service outages, additional UPS devices are deployed in the network infrastructure, and specialized UPS units support all servers. In the same way that layered protection enhances stability in immersive systems, this methodical and tiered approach increases reliability

The Executive Suite, Marketing, Sales, IT, Accounting, and Shipping/Receiving departments share thirty Windows 11 Pro workstations and five networked printers throughout the Office Work Area. CAT7 cabling is used to link each workstation back to the network, and orderly horizontal runs are routed from the server room into each department. Supported by a \$5,000 installation and termination allowance, this uncomplicated cabling topology reflects a design philosophy focused on ease of troubleshooting, accuracy, and simplicity.

This physical arrangement is supported by a Microsoft Active Directory tree that mirrors Techopoly's organizational structure. Centralized authentication, streamlined permissions, and easier administration are enabled by grouping users, devices, printers, and servers into distinct organizational units and role-based security groups. All things considered, this network architecture offers Techopoly a safe perimeter, robust and scalable server connectivity, a transparent access layer, and a clean distribution layer that supports each employee while planning for future expansion.